What is not returning to baseline?

Aegis REBOUND

A respiratory twin that chooses the next useful observation

Passive activity. Competing explanations. A limited measurement budget.

Proposed RECAP algorithm. Entirely synthetic; not clinically validated.

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One signal can fit several explanations

HYPOTHESIS 01

Ordinary recovery

HYPOTHESIS 02

Airway deterioration

HYPOTHESIS 03

Sensor drift

Raised respiratory rate is an observation. The explanation remains uncertain.

Static history meets changing observations

HISTORICAL EHR

Personal resting HR/RR, peak-flow best, prior attacks

+

DYNAMIC STREAM

Wearable HR/RR, steps, exposure, sparse peak flow

A personal state that changes after every observed sample

No future samples, hidden scenario labels, or simulator states enter inference.

RECAP: estimate, disagree, ask, update

01 / EHR-conditioned hypothesis bank

Personal baselines + history

02 / Nine continuous state filters

Obstruction / trend / debt / bias

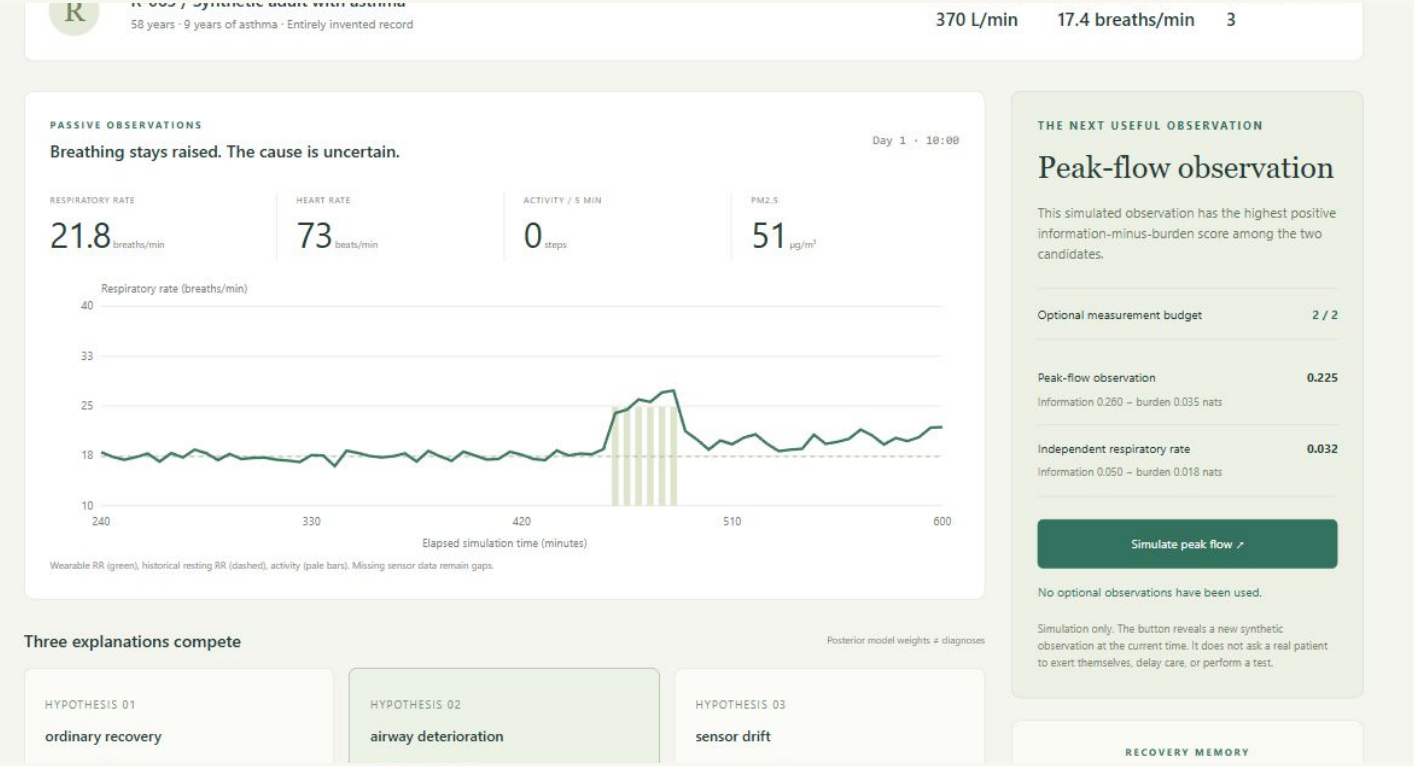
03 / Mixture forecast + calibrated risk index

24 five-minute future steps

04 / Expected information minus burden

Two optional probes; 120-minute cooldown

A measurement changes the virtual patient



Actual local app: R-065 at 10:00

Before: airway-family weight 81.2%; low-PEF event estimate 70.3%

After synthetic PEF: airway weight 97.1%; budget 1 of 2 remaining

Application capture. A synthetic observation updates the current-time state, not future truth.

The algorithm can also decide to wait

- Candidates: peak-flow ratio or independent respiratory rate.
- Nine-point Gaussian quadrature over possible outcomes.
- Score: expected risk/hypothesis entropy reduction minus assumed burden.
- Wait if no positive score, budget exhausted, or cooldown active.

A patient-disjoint, falsifiable experiment

+ 48 training / 16 calibration / 24 test patients
+ 904 held-out issue windows; 298 positive future windows
+ Fixed and adaptive policies share a maximum two additional probes.
+ Hand-specified filter dynamics; no claim of learned physiology.

Adaptive sensing used fewer extra observations

48

Fixed extra observations

Brier 0.0622

38

Adaptive extra observations

Brier 0.0593

Paired patient-mean Brier difference -0.00282

95% patient-bootstrap interval: -0.00536 to -0.00016

Complexity did not win every comparison

- Passive RECAP: AUROC 0.943; Brier 0.0643
- Single recovery setting: Brier 0.0633
- Wearable/context classifier: Brier 0.0574
- Recovery diversity did not improve event classification in this run.

No test-driven retuning was used to conceal this negative result.

A gap is a reason to withhold a forecast

- Freshness and completeness are checked before displaying inference.
- Sensor-gap example: 41.7% completeness, stale readings.
- No hypothesis ranking or event probability is shown.
- Bands elsewhere are uncalibrated model moments, not guarantees.

A proposed composition, with visible prior art

+ Asthma twins, recovery slowing and active sensing already exist.
+ The proposed contribution combines recovery-aware competing states with budgeted ambiguity resolution.
+ A targeted literature search does not establish global novelty or patentability.
+ The simulator cannot establish attack prediction or patient benefit.

Make the uncertainty inspectable

Team Aegis

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Working code, experiments, diagrams, docs, and recorded demonstration